OZONE : A TOXIC SAVIOR OF LIFE

AN INSIGHT of THE ACTUAL PHENOMENON

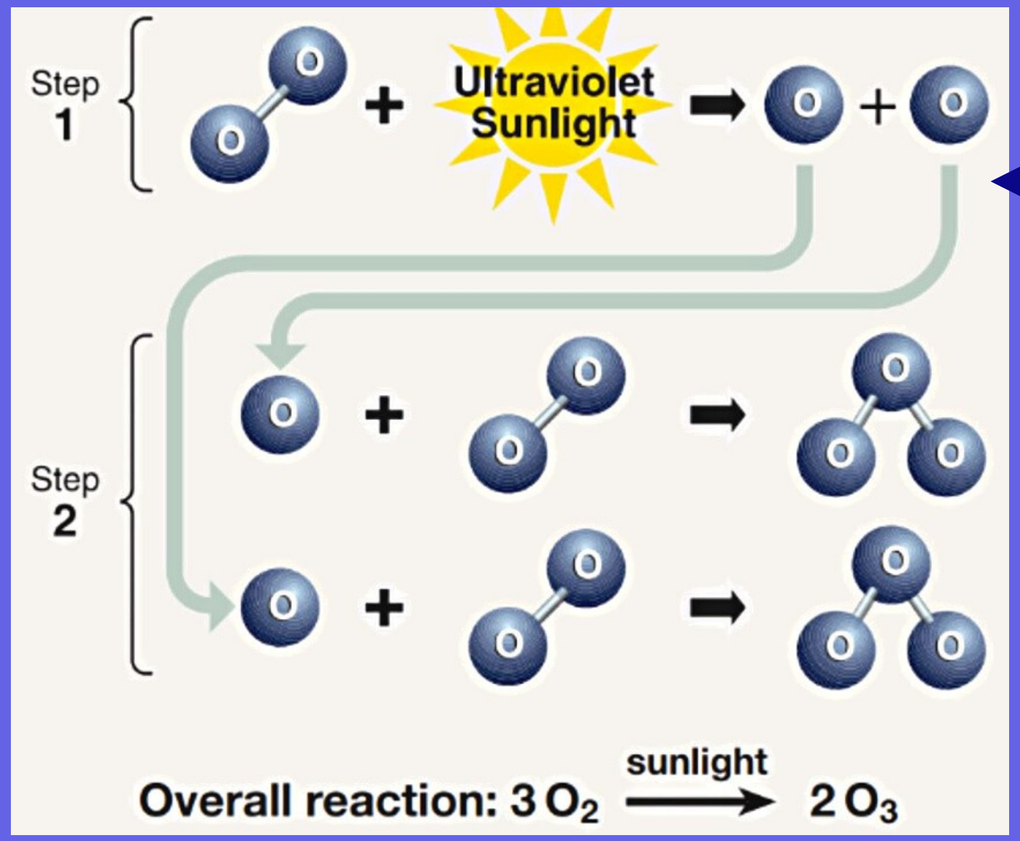
WHAT IS OZONE ?

Ozone is a pale blue colored gas, with a distinctively pungent smell. Ozone - O_3 (trioxygen allotrope) is an inorganic molecule which is much less stable than O_2 (dioxygen allotrope) of O (Oxygen, 21% of atmosphere).

O_3 breaks down in the lower atmosphere as O_2.



NATURAL CREATION OF OZONE



Largest Ozone-production occurs in the tropical-stratosphere

Stratospheric Ozone Production

In presence of solar ultraviolet radiations (sunlight), in steps, with oxygen (O_2 , 21% of atmosphere), chemical reactions take place to form ozone molecule, O_3



Why is **OZONE LAYER** so Important ?

STRATOSPHERIC OZONE - IMPORTANCE

Fauna & Human

Control UVB-entry into earth atmosphere; laboratory & epidemiologist found that:-

- UVB causes **Non-Melanoma Skin Cancer.**
- Promotes Malignant **Melanoma Development.**
- Is linked with the **Development of Cataracts** (a clouding of eye's lens).

Flora

Plants have abilities to adapt to high UVB-levels, still they're directly affected ---

- **Indirect UVB induced changes-** in **plant form, nutrient-distribution, developmental-phases timing, secondary metabolism.**
- Effects **plant's competitive balance, herbivory, plant-diseases & biogeochemical changes.**

Marine Ecosystems

Phytoplankton's (foundation of aquatic food webs) productivity being limited to euphotic zone, sunlight having UVB Radiation affects their **orientation & motility** , causing threat to aquatic lives.

UVB-Radiation causes damage to **Early-Developmental stages** of many marine animals, e.g. reduced reproductive capacity, impaired larval development.

Bio-geochemical Cycles

UVB affects **terrestrial & aquatic Biogeochemical Cycles**, altering both sources & sinks of Greenhouse & chemically important Trace-gases **(e.g. CO2, CO, COS, O 3, etc.**)

These potential changes contributing to biosphere-atmosphere feedbacks, mitigates or amplifies the atmospheric concentrations of these gases.

Materials

Synthetic Polymers, natural bio-polymers, commercially sold materials are highly damaged and their quality & longevity is affected by UVB Radiation.

WHY IS Ground level OZONE LIFE-THREATENING ?

TROPOSPHERIC OZONE EXPOSURE

Depending on the level of 'Troposhepric Ozone Exposure' it is detrimental in several ways :

- Hinders vigorous and deep breathing; pain while breathing deeply
- Inflammation & irritation of the tissue-linings of human-airways
- Lungs turn more susceptible to infection
- Aggravate lung diseases like- asthma, emphysems & chronic bronchitis
- Increases frequency of Asthma-Attacks
- Ozone causes airway-muscles to constrict, trapping air in the alveoli- leading to wheezing & shortness of breath



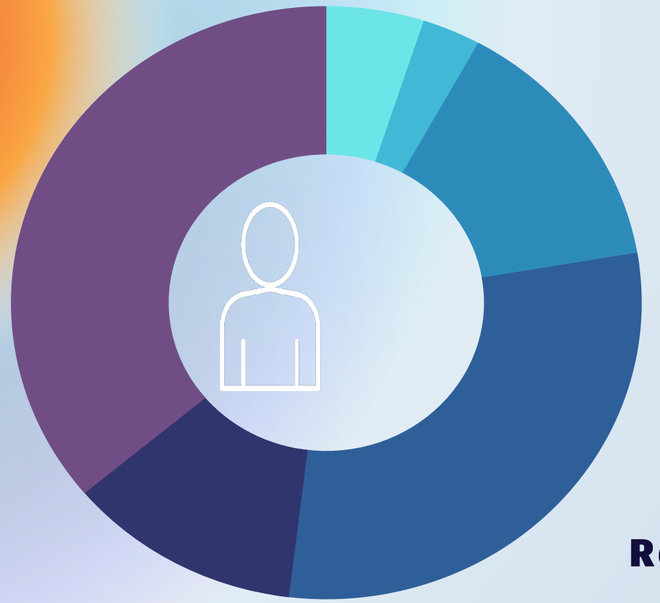
HUMAN-ACTIONS BEHIND OZONE-LAYER CHANGES

ANTHROPOGENIC OZONE-LAYER CHANGES

Human Civilization Growth has always prioritized comfort over Environmental-Balance : this self-centered attitude can actually threaten the future-generations and survival of other species.

Different Industrial Emissions, household use of refrigerators producing **CFCs**, automobile engines generating harmful **Greenhouse-Gases (CFCs & HCFCs)** that form **Chlorine & Bromine free-radicals** that are mainly responsible for breaking down thus destroying the ozone-layer.

O_3 to O_2

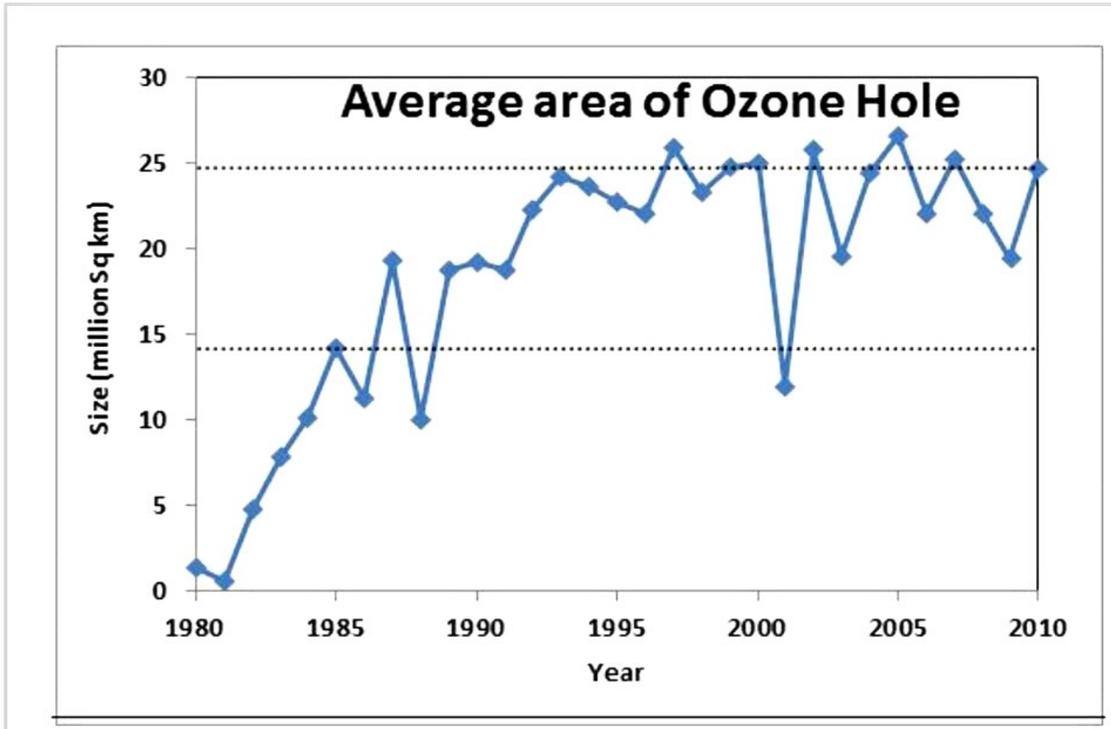


GRAPH

Extreme use of CFC-gas, petrol (producing CO2) destroying ozone-layer.

Since 1979~1980 through to the early 1990s, stratospheric ozone-concentrations fell to the concerning ozone-hole level of **100 DU**; which continued approximately **around 100 DU (or below)** for several following decades...

DU(Dobson Units): Measurement unit for ozone-layer thickness in an air-column from ground to atmosphere-top



Southern hemisphere ozone hole area.

OZONE-HOLE :

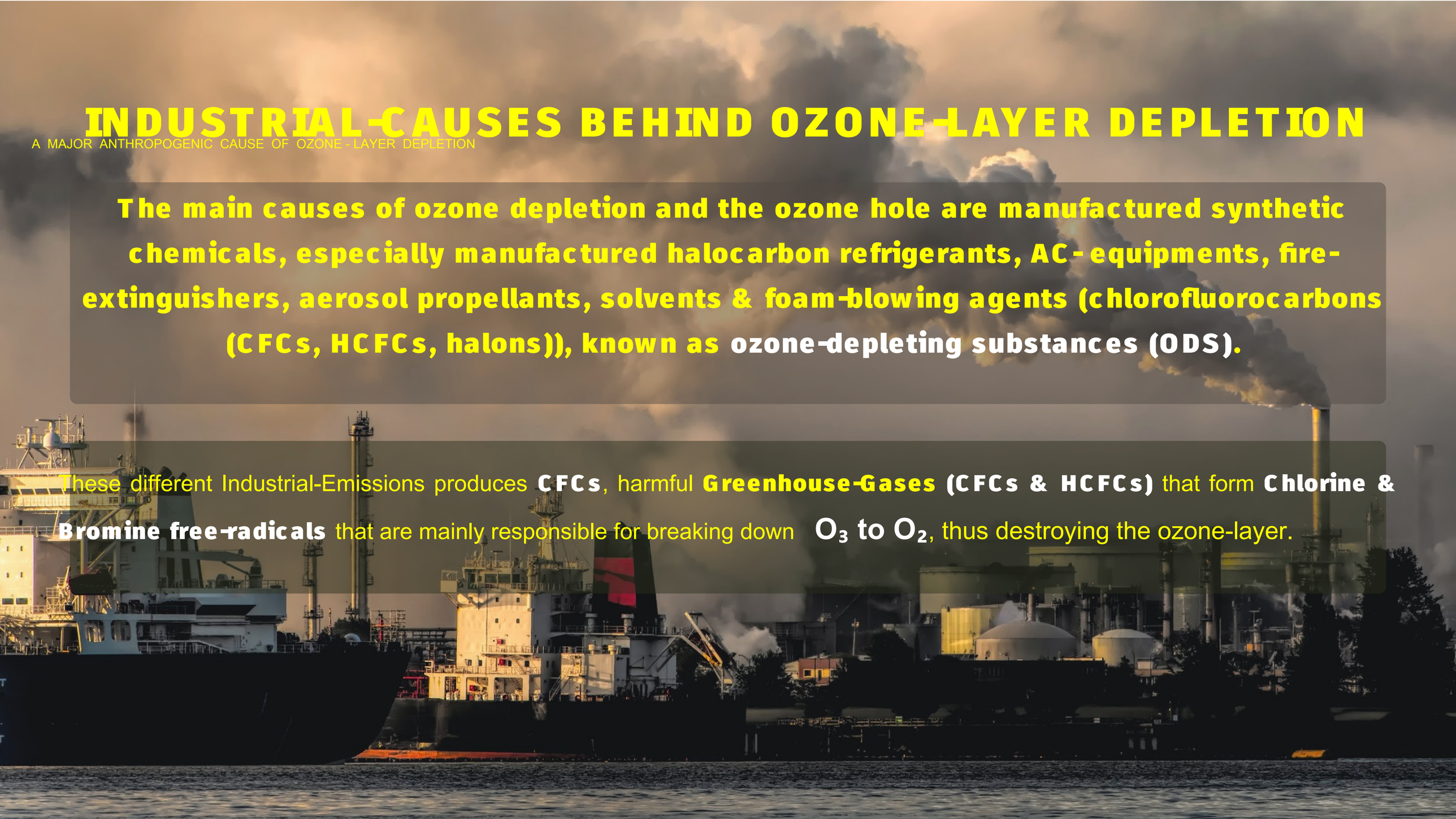
Area having less than 220 DU of ozone (concentration) in the overhead column (i.e., between the ground and space)

The graph compares the size of 'Southern-Hemisphere ozone-hole' over a **20 yrs. period time, from 1980 - 2010**; where each year, **at spring, ozone hole turns largest**

INDUSTRIAL-CAUSES BEHIND OZONE-LAYER DEPLETION

A MAJOR ANTHROPOGENIC CAUSE OF OZONE-LAYER DEPLETION

The main causes of ozone depletion and the ozone hole are manufactured synthetic chemicals, especially manufactured halocarbon refrigerants, AC-equipments, fire-extinguishers, aerosol propellants, solvents & foam-blowing agents (chlorofluorocarbons (CFCs, HCFCs, halons)), known as ozone-depleting substances (ODS).



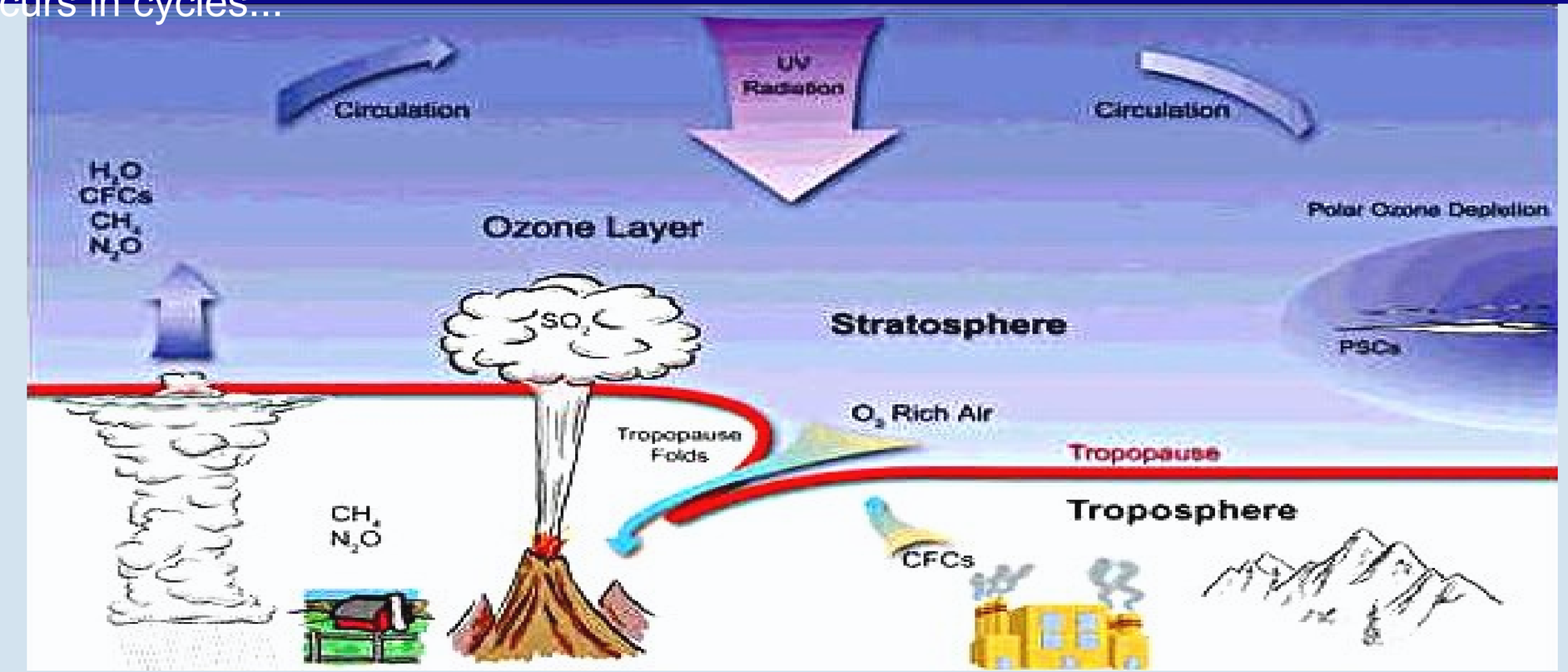
These different Industrial-Emissions produces **CFCs**, harmful **Greenhouse-Gases (CFCs & HCFCs)** that form **Chlorine & Bromine free-radicals** that are mainly responsible for breaking down O_3 to O_2, thus destroying the ozone-layer.

O_3 → O_2

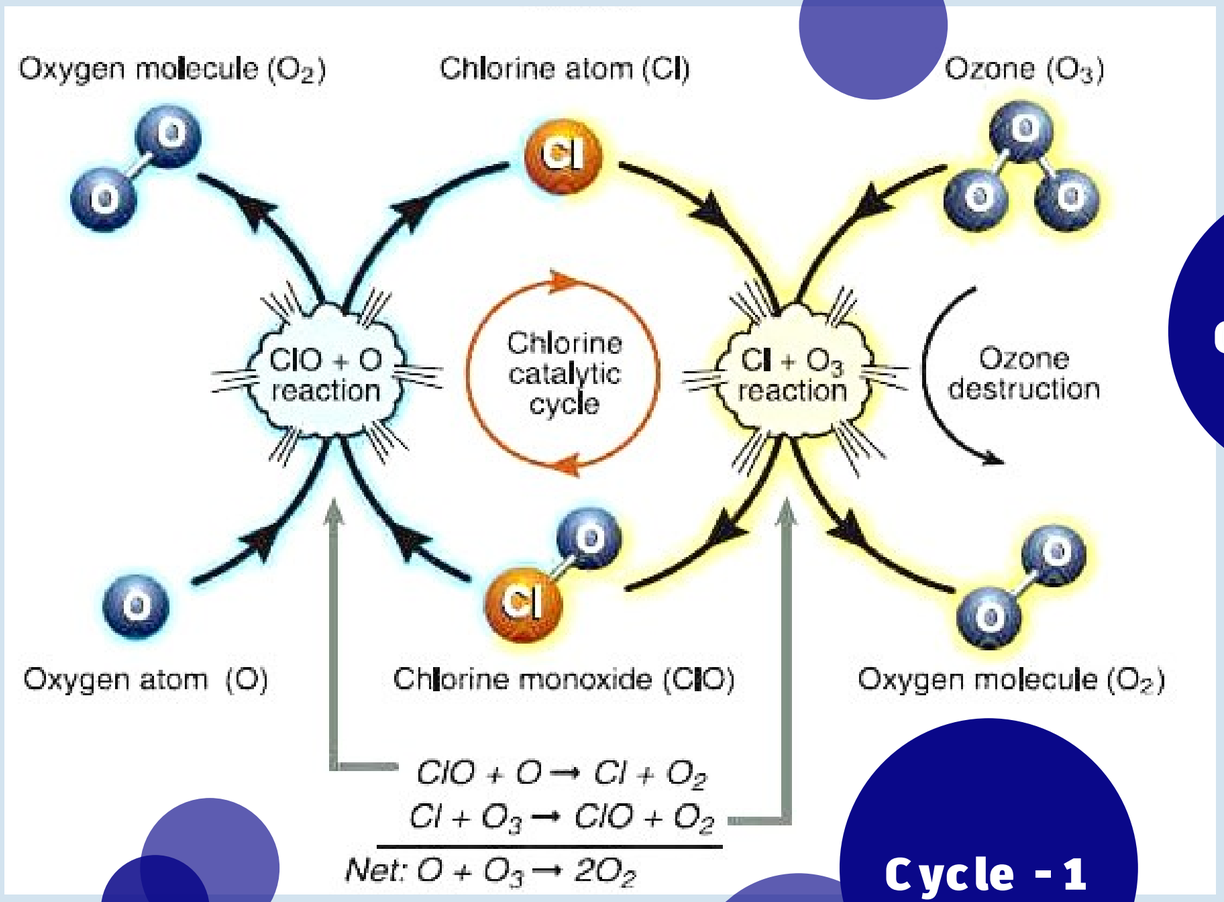
STRATOSPHERIC OZONE - LAYER DEPLETION

Human activities cause the emission of halogen source gases that contain chlorine and bromine atoms. These emissions into the atmosphere ultimately lead to **stratospheric ozone depletion**. **The source gases that contain only carbon, chlorine, and fluorine are called "chlorofluorocarbons," usually abbreviated as CFCs.**

Once into the stratosphere, through photo-dissociation, catalytic-breakdown of ozone (**O_3 to O_2**) occurs in cycles...



Cycles of Ozone-Layer Depletion



Cycle - 1

Cycle - 2

$$ClO + ClO \rightarrow (ClO)_2$$
$$(ClO)_2 + \text{sunlight} \rightarrow ClOO + Cl$$
$$ClOO \rightarrow Cl + O_2$$
$$2(Cl + O_3 \rightarrow ClO + O_2)$$

Net: $2O_3 \rightarrow 3O_2$

Cycle - 3

$$ClO + BrO \rightarrow Cl + Br + O_2$$

or

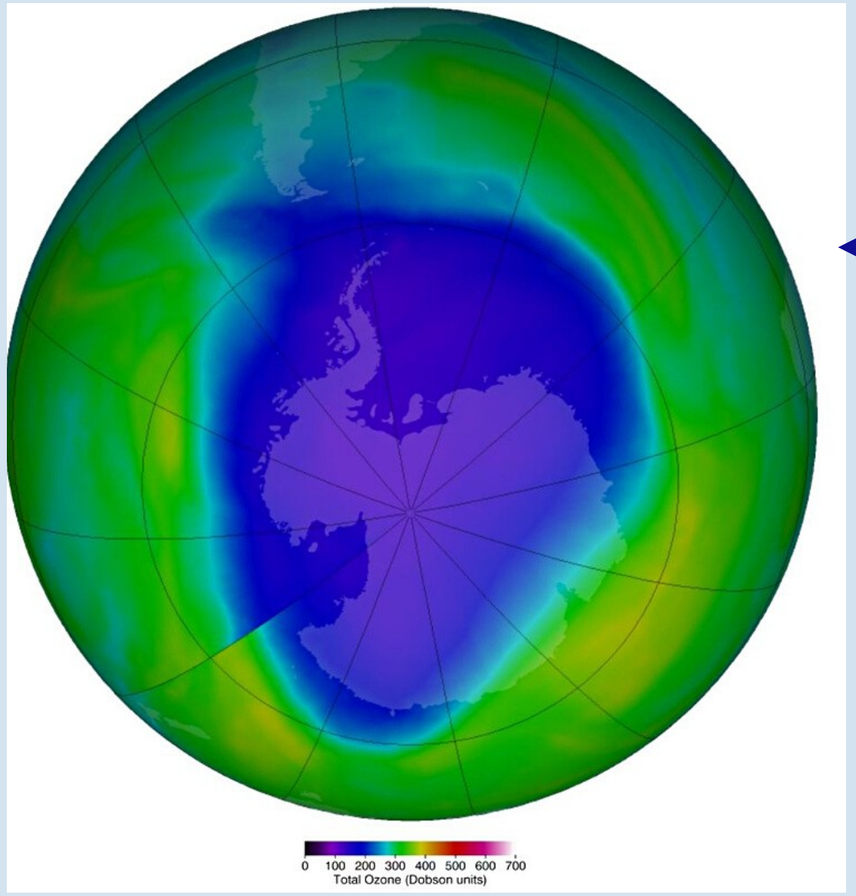
$$ClO + BrO \rightarrow BrCl + O_2$$
$$BrCl + \text{sunlight} \rightarrow Cl + Br$$

$$Cl + O_3 \rightarrow ClO + O_2$$
$$Br + O_3 \rightarrow BrO + O_2$$

Net: $2O_3 \rightarrow 3O_2$

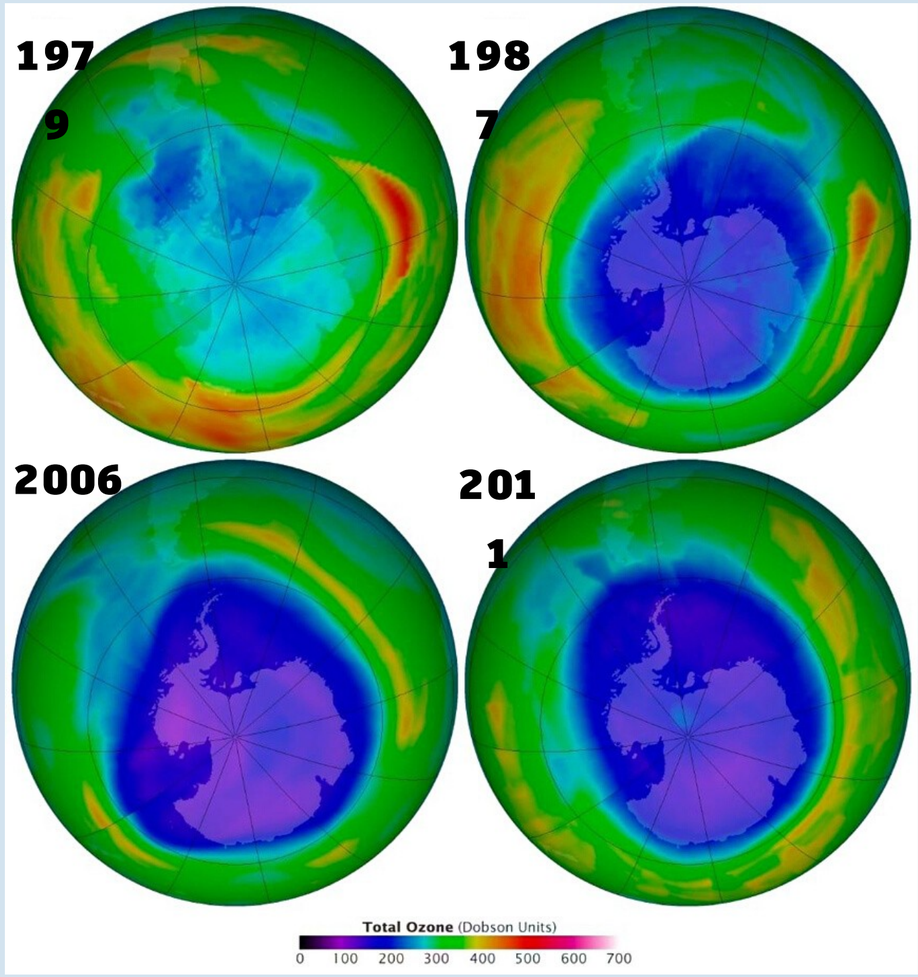
THE OZONE-HOLE

A MOST RECENT DATA



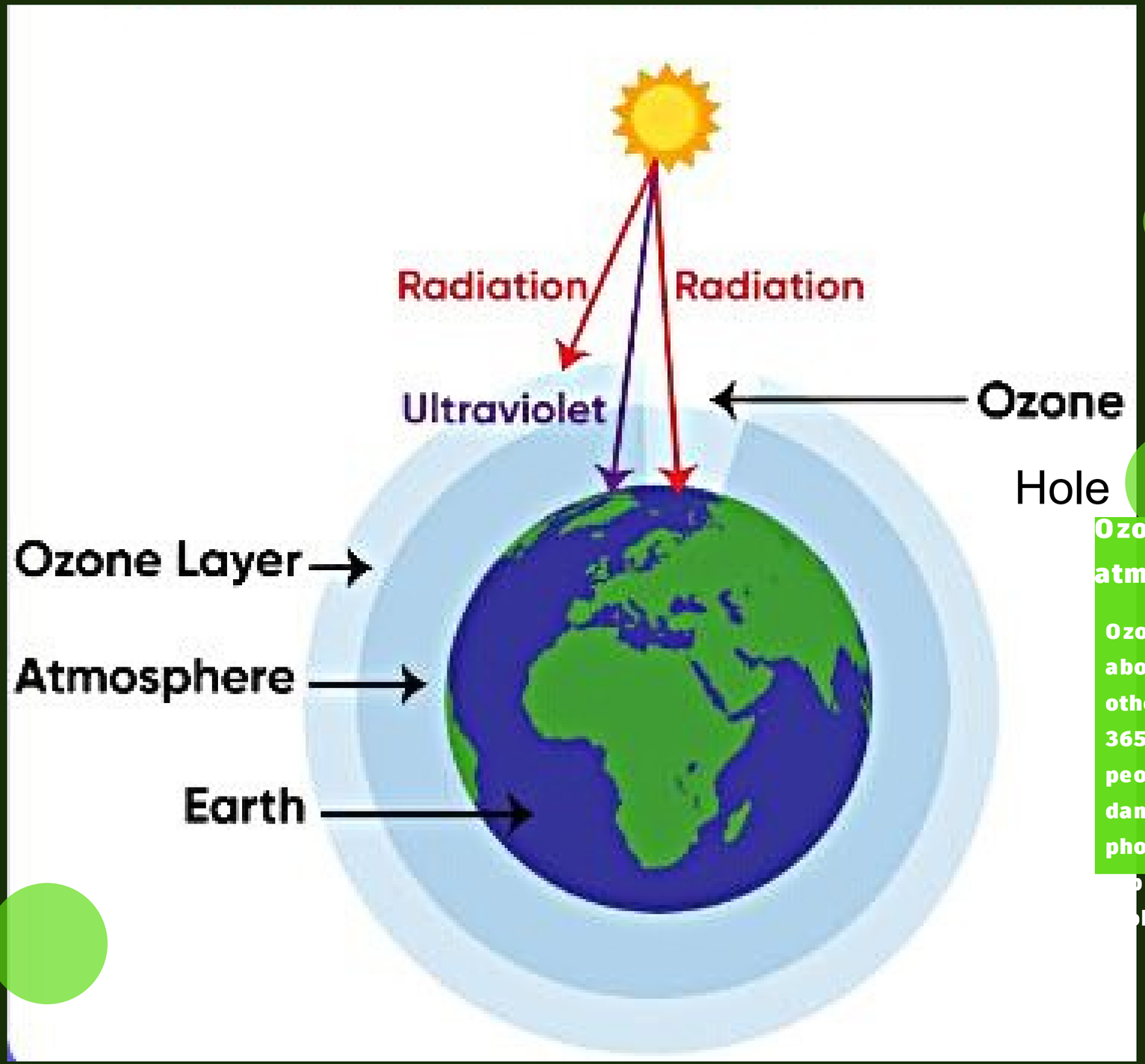
3 November, 2022

False-color view of total ozone over the Antarctic pole. The purple and blue colours are where there is the least ozone, and the yellows and reds are where there is more ozone



OZONE-HOLE CHANGING





OZONE-HOLE EFFECTS

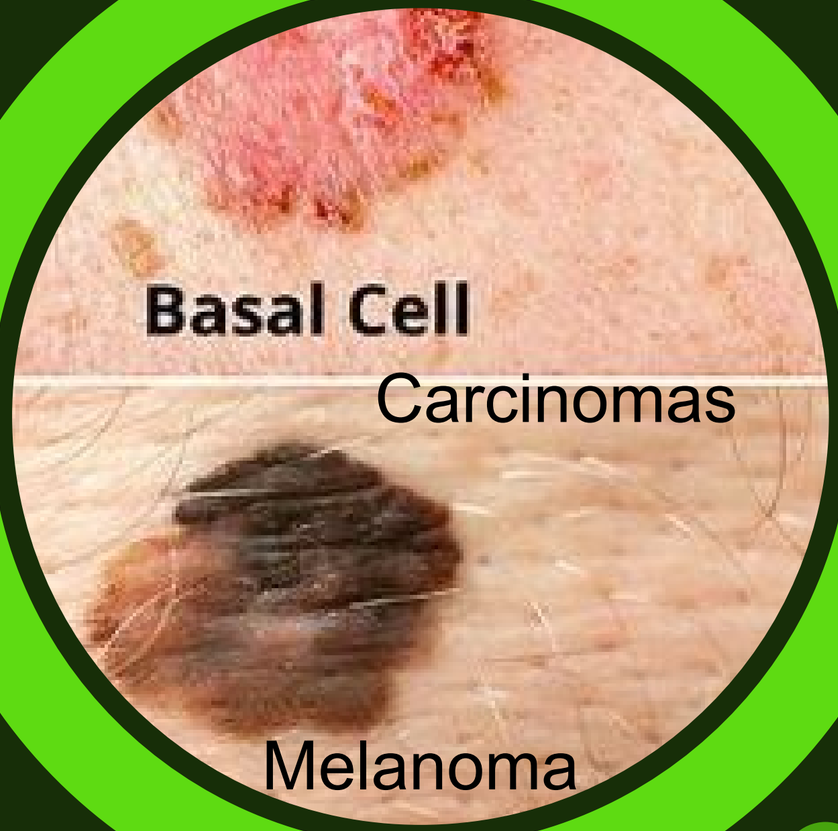
Ozone Hole allowing UVB Radiation into the atmosphere, on the earth surface

Ozone protects us from destructive solar radiation when it's high above Earth in the stratosphere. Closer to the ground, though, it has other lasting impacts. Ozone at the surface was estimated to cause 365,000 deaths globally in 2019 by damaging the lungs of vulnerable people, such as young children and those with asthma. Similarly, it damages the breathing systems of plants – their ability to photosynthesize – reducing plant growth and crop yields. And at the p of the troposphere, it's a potent greenhouse gas, increasing bal temperatures.

STRATOSPHERIC OZONE DECRES & TROPOSPHERIC OZONE INCREASE

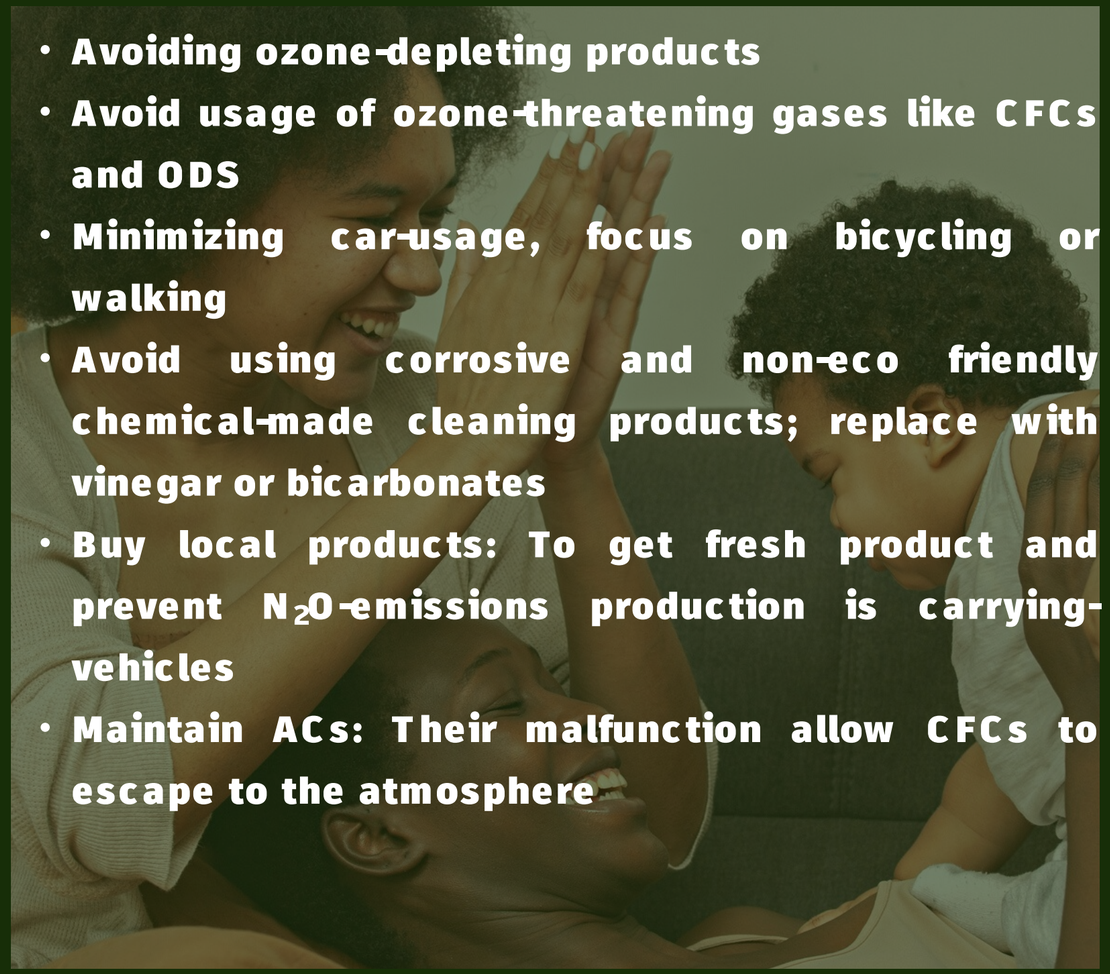
THREATS OF OZONE-IMBALANCE

- Human Health: positive - increases vitamin-D production
  negative - sunburn, skin-cancer, cataracts
- Basal & Squamous Cell Carcinomas: causes pyridine molecules to form dimers, resulting in transcription errors while DNA-replication
- Malignant Melanoma: 19% increase (men) & 16% increase (women) in melanomas
- Cortical Cataracts
- Tropospheric-Ozone can be fatal for its highly oxidizing properties for people with asthma and breathing-difficulties
- Animals: Epidermal damages of whales in the Gulf of California associated with acute & severe sunburn have been found
- Crops: Plant growth patterns disturbed & cyanobacteria is sensitive to UVB-Radiation



BALANCED SUSTENANCE OF HUMAN & OTHER SPECIES

COMPORT COMPLYING WITH OZONE-PROTECTION

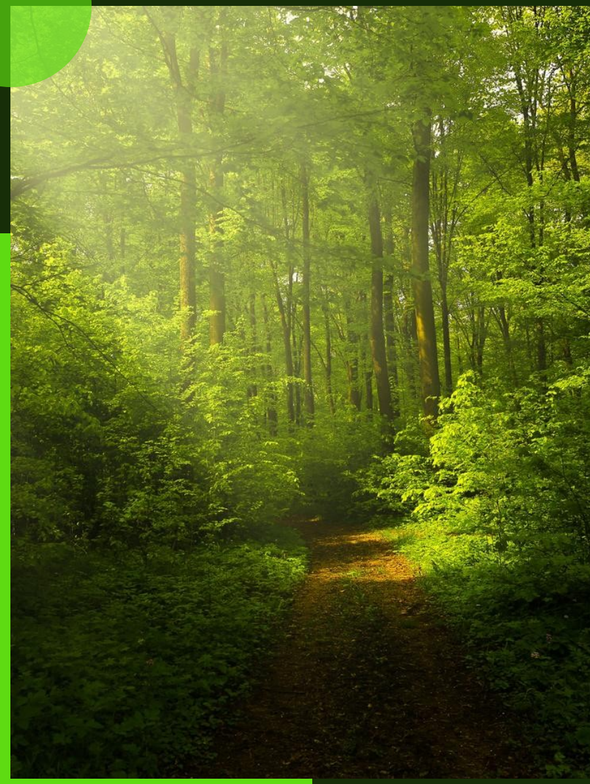


- Avoiding ozone-depleting products
- Avoid usage of ozone-threatening gases like CFCs and ODS
- Minimizing car-usage, focus on bicycling or walking
- Avoid using corrosive and non-eco friendly chemical-made cleaning products; replace with vinegar or bicarbonates
- Buy local products: To get fresh product and prevent N_2O-emissions production is carrying-vehicles
- Maintain ACs: Their malfunction allow CFCs to escape to the atmosphere



CORONA PANDEMIC OBSERVATIONS

CERTAIN INDUSTRIAL-EMISSIONS CONTRIBUTE TO O_3 - FORMATION





Written on June 9, 2021, NASA found -
"Local lockdowns brought fast global ozone-reductions

Emissions of nitrogen oxides (NOx) – which create ozone, a danger to human health and to climate – decreased 15% globally, with local reductions as high as 50%

As a result of the lower NOx emissions, by June 2020, global ozone levels had dropped to a level that policy makers thought would take at least 15 years to reach by conventional means, such as regulations.



INTERNATIONAL PROTOCOLS

INTERNATIONAL PROTOCOLS ADOPTED FOR OZONE PROTECTION

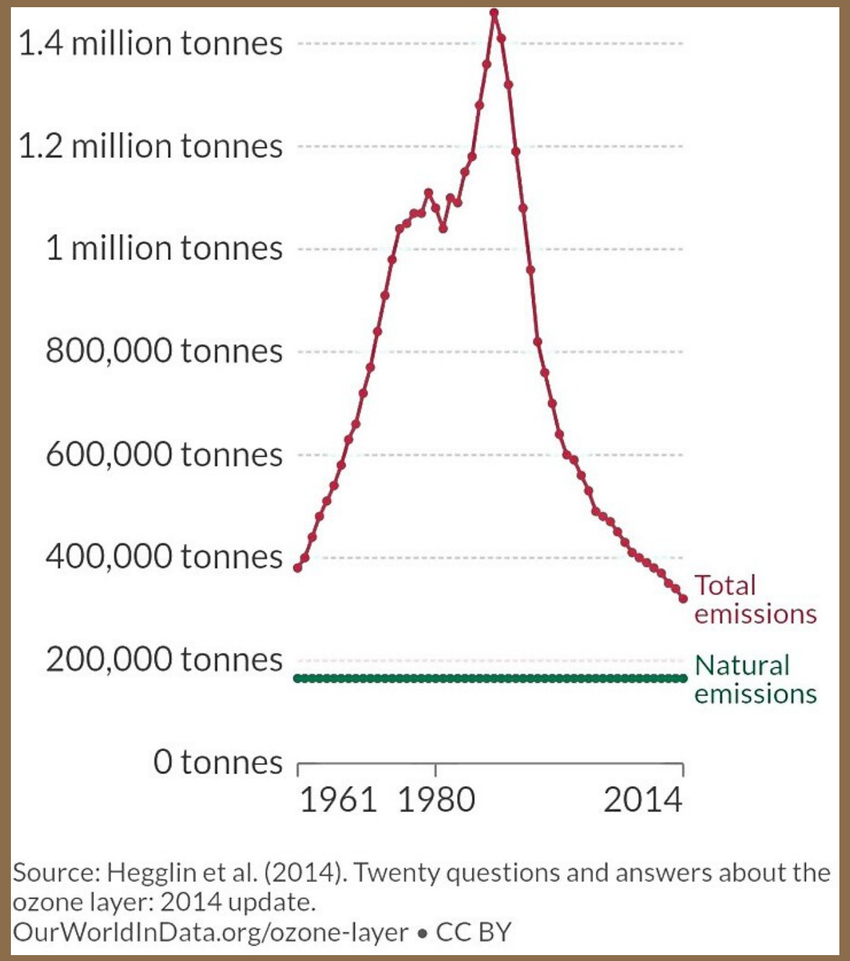
- The Vienna Convention - 1985
- The Montreal Protocol - 1987

Amendments to the Montreal Protocols:

- **The London Amendment - 1990**
- **The Copenhagen Amendment - 1992**
- **The Montreal Amendment - 1997**
- **The Beijing Amendment - 1999**
- **The Kigali Amendment - 2016**



EMISSIONS : RISE & FALL



In the chart beside,
emissions of Ozone-depleting substances are seen from 1960 onwards, measured in tons of CFC-11 equivalents.

CFC-11 is a standardized measure to normalize the sum of a range of ozone-depleting substances, to a value equivalent to their potential to deplete ozone.

Here, a clear growth-peak-reduction is seen with also a rapid rise by 3 folds from 1960s to 1980s.

By 2010, emissions returned to 1960 levels a result of various strict international regulatory agreements



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- https://ww2.arb.ca.gov/resoucesozone-and-health
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- https://ourworldindata.org/grapher/ozone-depleting-substance-emissions



THANK YOU